TASK SPECIFICATION: P5-09 PHASE P5 — SMART PARKING & ALPR COMPUTER VISION GATE

ALPR Camera Scanner & IoT Sensor Simulator UI

Git Branch: `feature/P5-09-gate-sim-ui` · Target Pull Request: feature/P5-09-gate-sim-ui

Core Objective & Technical Scope:
Interactive presentation simulator demonstrating gate camera OCR recognition, distance slider sensor injection, and hardware diagnostics.

Priority Level	Low	Complexity Rating	M
Execution Status	Backlog 📋	Target PR / Commit	`feature/P5-09-gate-sim-ui`
Upstream Dependencies	P5-05		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Build GateScannerPreview demonstrating OCR plate framing and barrier animation.
2. Build SimulatorPanel with ultrasonic sensor distance sliders and event triggers.
3. Integrate subtle audio feedback synthesizer for successful badge scans.
4. Wire simulator events to backend gate decision endpoint.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/src/features/parking/gate/*`
- `simulator/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Demonstrates full entry/exit pipeline with realistic visual telemetry.
- Audio synthesizer plays feedback without blocking the UI thread.

VERIFICATION & TEST COVERAGE STRATEGY

- GateScanner component tests
- Simulator interaction tests